

# CoppeTeX quick reference

Class `coppe`, v4.1 — every option, command and environment

This is the short companion to `coppe.pdf`, the full manual, which is written in Portuguese. A COPPE thesis may be written in Portuguese, English or Spanish (CEPG Resolution 302/2024, art. 57); this sheet is for whoever writes it in English.

The **Where** column says where a command may appear: PRE means the preamble, before `\begin{document}`; DOC means after `\begin{document}`, and, for everything that feeds the cover, *before* `\maketitle`.

**Three things that surprise people.** The document data goes *inside* `document`, not in the preamble. The bibliography is `biblatex` with `biber`, not `BibTeX`. And `\frontmatter`/`\mainmatter` are what switch folio numbering on, so leaving them out gives the wrong page numbers.

| Command                                           | Example                                      | Where | What it does                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------|----------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CLASS OPTIONS — work type, one is required</b> |                                              |       |                                                                                                                                                                                                                                                                                                                 |
| <code>msc</code>                                  | <code>\documentclass[msc]{coppe}</code>      | PRE   | M.Sc. dissertation.                                                                                                                                                                                                                                                                                             |
| <code>dsc</code>                                  | <code>\documentclass[dsc]{coppe}</code>      | PRE   | D.Sc. thesis.                                                                                                                                                                                                                                                                                                   |
| <code>mscexam</code>                              | <code>\documentclass[mscexam]{coppe}</code>  | PRE   | M.Sc. qualifying exam. No CAPES sheet, no catalogue card, no reference above the abstract.                                                                                                                                                                                                                      |
| <code>dsccexam</code>                             | <code>\documentclass[dsccexam]{coppe}</code> | PRE   | D.Sc. qualifying exam. Same.                                                                                                                                                                                                                                                                                    |
| <code>mscsem</code>                               | <code>\documentclass[mscsem]{coppe}</code>   | PRE   | M.Sc. seminar, a requirement of some Programas and <i>not</i> a qualifying exam. Same three omissions: it is not deposited.                                                                                                                                                                                     |
| <b>CLASS OPTIONS — writing language</b>           |                                              |       |                                                                                                                                                                                                                                                                                                                 |
| <code>brazilian</code>                            | <code>[brazilian,dsc]</code>                 | PRE   | Portuguese. The default; needs no writing.                                                                                                                                                                                                                                                                      |
| <code>english</code>                              | <code>[english,dsc]</code>                   | PRE   | English. The foreign abstract switches to Portuguese.                                                                                                                                                                                                                                                           |
| <code>spanish</code>                              | <code>[spanish,dsc]</code>                   | PRE   | Spanish. Needs <code>coppe-lang-spanish.def</code> and <code>spanish-coppe.lbx</code> . Three abstracts.                                                                                                                                                                                                        |
| <code>french</code>                               | <code>[french,dsc]</code>                    | PRE   | French. <b>Not</b> a permitted thesis language; ships as a demonstration.                                                                                                                                                                                                                                       |
| <code>italian</code>                              | <code>[italian,dsc]</code>                   | PRE   | Italian. Same.                                                                                                                                                                                                                                                                                                  |
| <b>CLASS OPTIONS — the rest</b>                   |                                              |       |                                                                                                                                                                                                                                                                                                                 |
| <code>numbers</code>                              | <code>[dsc,numbers]</code>                   | PRE   | Numeric citations [1] instead of author-date; list in citation order.                                                                                                                                                                                                                                           |
| <code>pdfa</code>                                 | <code>[dsc,pdfa]</code>                      | PRE   | PDF/A-2b output, required for deposit (2.2d): the deposit is digital only, and PDF/A (ISO 19005) keeps the file self-contained — embedded fonts, sRGB colour, XMP metadata, nothing external. Off by default; <code>example.tex</code> turns it on, and turning it on early is recommended. Check with veraPDF. |
| <code>comserifa</code>                            | <code>[dsc,comserifa]</code>                 | PRE   | Serif family. Sans is the default since v4.1.                                                                                                                                                                                                                                                                   |

| Command                           | Example                                                     | Where | What it does                                                                                                                                                                                                                                                |
|-----------------------------------|-------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>semserifa</code>            | --                                                          | PRE   | Accepted, does nothing. What it asked for became the default.                                                                                                                                                                                               |
| <code>semlinks</code>             | <code>[dsc,semlinks]</code>                                 | PRE   | Links lose colour and frame; they stay clickable. For printing.                                                                                                                                                                                             |
| <code>doublespacing</code>        | <code>[dsc,doublespacing]</code>                            | PRE   | Double spacing instead of one-and-a-half (2.2c).                                                                                                                                                                                                            |
| <code>twoside</code>              | <code>[dsc,twoside]</code>                                  | PRE   | Mirrored margins, for printing and binding.                                                                                                                                                                                                                 |
| <code>oneside</code>              | <code>[dsc,oneside]</code>                                  | PRE   | One-sided, 3 cm left margin on every sheet. The default.                                                                                                                                                                                                    |
| <code>orientadorexamina</code>    | <code>[dsc,orientadorexamina]</code>                        | PRE   | Put advisors back on the approval sheet. Off by default: the sheet records who examined.                                                                                                                                                                    |
| <code>coorientador</code>         | <code>[dsc,coorientador]</code>                             | PRE   | List coadvisors on the abstract pages too.                                                                                                                                                                                                                  |
| <code>rascunhoficha</code>        | <code>[dsc,rascunhoficha]</code>                            | PRE   | Draft catalogue card while writing. <b>Never valid for deposit.</b>                                                                                                                                                                                         |
| <code>resumosemreferencia</code>  | <code>[dsc,resumosemreferencia]</code>                      | PRE   | Drop the work's own reference from the top of the three abstracts. It is printed by default (3.1.2.1.4, Annexes E and F).                                                                                                                                   |
| <code>listasnosumario</code>      | <code>[dsc,listasnosumario]</code>                          | PRE   | Put the pre-textual lists back in the table of contents.                                                                                                                                                                                                    |
| <code>assinaturas</code>          | --                                                          | PRE   | Accepted, does nothing. Signature rules are gone: deposit is digital.                                                                                                                                                                                       |
| <code>numeraisromanos</code>      | <code>[dsc,numeraisromanos]</code>                          | PRE   | Roman numerals on pre-textual sheets. <b>Against 2.7</b> ; legacy only.                                                                                                                                                                                     |
| <code>morewrites</code>           | <code>[dsc,morewrites]</code>                               | PRE   | A missing <code>morewrites</code> package becomes an error, not a warning.                                                                                                                                                                                  |
| <code>semmorewrites</code>        | <code>[dsc,semmorewrites]</code>                            | PRE   | Do not load <code>morewrites</code> . Saves about half a second per pdf <sub>l</sub> atex pass (Overleaf free plan). Safe only without an index, several indexes, a glossary or nomenclature: the class stops with a clear error if the 16 streams run out. |
| <b>PREAMBLE</b>                   |                                                             |       |                                                                                                                                                                                                                                                             |
| <code>\addbibresource</code>      | <code>\addbibresource{thesis.bib}</code>                    | PRE   | Declare the bibliography database. With the <code>.bib</code> extension.                                                                                                                                                                                    |
| <code>\makelosymbols</code>       | <code>\makelosymbols</code>                                 | PRE   | Turn on collection of the list of symbols. Without it, <code>\syml</code> records nothing.                                                                                                                                                                  |
| <code>\makeloabbreviations</code> | <code>\makeloabbreviations</code>                           | PRE   | Same for the list of abbreviations.                                                                                                                                                                                                                         |
| <code>\newcoppefloat</code>       | <code>\newcoppefloat{map}{Map}{List of Maps}</code>         | PRE   | Declare your own float: caption on top, <code>\source</code> below, own list.                                                                                                                                                                               |
| <code>\newsigla</code>            | <code>\newsigla{ufrj}{UFRJ}{Universidade Federal...}</code> | PRE   | Declare an acronym for <code>\sigla</code> .                                                                                                                                                                                                                |
| <code>\usecoppelanguage</code>    | <code>\usecoppelanguage{french}</code>                      | PRE   | Load one more language pack, for quotations in a third language.                                                                                                                                                                                            |
| <code>\graphicspath</code>        | <code>\graphicspath{{figures/}}</code>                      | PRE   | Where your figures live. Standard L <sup>A</sup> T <sub>E</sub> X.                                                                                                                                                                                          |

| Command                                                                                     | Example                                                | Where | What it does                                                                                                   |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------|
| <code>\freeconfig</code>                                                                    | <code>\freeconfig{Dr.}{...}{...}{...}{...}</code>      | PRE   | Override the work type entirely. Escape hatch; do not use for a normal thesis.                                 |
| <b>IDENTIFICATION — after <code>\begin{document}</code>, before <code>\maketitle</code></b> |                                                        |       |                                                                                                                |
| <code>\title</code>                                                                         | <code>\title{Titulo da tese}</code>                    | DOC   | The Portuguese title.                                                                                          |
| <code>\foreigntitle</code>                                                                  | <code>\foreigntitle{Thesis title}</code>               | DOC   | The English title.                                                                                             |
| <code>\titlein</code>                                                                       | <code>\titlein{spanish}{Titulo}</code>                 | DOC   | The title in any other language.                                                                               |
| <code>\subtitle</code>                                                                      | <code>\subtitle{Um subtítulo}</code>                   | DOC   | Optional subtitle, Portuguese. Printed after a colon, in ordinary case.                                        |
| <code>\foreignsubtitle</code>                                                               | <code>\foreignsubtitle{A subtitle}</code>              | DOC   | The same, English.                                                                                             |
| <code>\subtitlein</code>                                                                    | <code>\subtitlein{spanish}{...}</code>                 | DOC   | The same, any language.                                                                                        |
| <code>\author</code>                                                                        | <code>\author{All given names}{Surname}</code>         | DOC   | Two arguments: the catalogue card has to invert the name.                                                      |
| <code>\advisor</code>                                                                       | <code>\advisor{Ana}{Lima}{D.Sc.}{UFRJ}</code>          | DOC   | Name, surname, degree, institution. Repeat per advisor. Optional first argument is the treatment.              |
| <code>\coadvisor</code>                                                                     | <code>\coadvisor{Bia}{Sousa}{Ph.D.}{UFF}</code>        | DOC   | Same four arguments.                                                                                           |
| <code>\examiner</code>                                                                      | <code>\examiner{Carlos Dias}{D.Sc.}{USP}</code>        | DOC   | Name and surname in one argument, then degree and institution.                                                 |
| <code>\department</code>                                                                    | <code>\department{PESC}</code>                         | DOC   | The Programa's acronym. The class knows all thirteen.                                                          |
| <code>\date</code>                                                                          | <code>\date{09}{2026}</code>                           | DOC   | Month and year of deposit, as numbers. Not the defence date.                                                   |
| <code>\dataaprovacao</code>                                                                 | <code>\dataaprovacao{15 de setembro de 2026}</code>    | DOC   | Defence date, in words. Without it the sheet reads “a ser determinada”.                                        |
| <code>\keyword</code>                                                                       | <code>\keyword{Formal methods}</code>                  | DOC   | One call per keyword, main language.                                                                           |
| <code>\foreignkeyword</code>                                                                | <code>\foreignkeyword{Métodos formais}</code>          | DOC   | One call per keyword, foreign language.                                                                        |
| <code>\braziliankeyword</code>                                                              | <code>\braziliankeyword{Métodos formais}</code>        | DOC   | Portuguese keywords when the work is not in Portuguese.                                                        |
| <code>\volumes</code>                                                                       | <code>\volumes{3}</code>                               | DOC   | How many volumes. Nothing is printed when there is one.                                                        |
| <code>\volume</code>                                                                        | <code>\volume{1}</code>                                | DOC   | Which volume this is.                                                                                          |
| <b>TITLE PAGE AND CAPES SHEET</b>                                                           |                                                        |       |                                                                                                                |
| <code>\areaconcentracao</code>                                                              | <code>\areaconcentracao{Engenharia de Sistemas}</code> | DOC   | Area of concentration. On three sheets.                                                                        |
| <code>\linhapesquisa</code>                                                                 | <code>\linhapesquisa{Engenharia de Dados}</code>       | DOC   | Research line, on the title page. Label follows the main language. Empty is fine.                              |
| <code>\tipoproducao</code>                                                                  | <code>\tipoproducao{bibliografica}</code>              | DOC   | One of <code>bibliografica</code> , <code>artistica</code> , <code>tecnologica</code> , <code>tecnica</code> . |
| <code>\projetovinculado</code>                                                              | <code>\projetovinculado{sim}</code>                    | DOC   | <code>sim</code> or <code>nao</code> : is there a linked research project?                                     |
| <code>\nomeprojeto</code>                                                                   | <code>\nomeprojeto{Nome do projeto}</code>             | DOC   | Its name, if there is one.                                                                                     |
| <code>\agenciafomento</code>                                                                | <code>\agenciafomento{Conselho...}{CNPq}</code>        | DOC   | Funding agency, full name and acronym. One call each.                                                          |
| <code>\fichacatalografica</code>                                                            | <code>\fichacatalografica{ficha.pdf}</code>            | DOC   | The official catalogue card, as a PDF from the SiBI generator.                                                 |
| <b>STRUCTURE</b>                                                                            |                                                        |       |                                                                                                                |
| <code>\maketitle</code>                                                                     | <code>\maketitle</code>                                | DOC   | Typesets cover, title page and CAPES sheet. After all the data.                                                |

| Command                         | Example                                                | Where | What it does                                                                                                                                                                                                                             |
|---------------------------------|--------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>\frontmatter</code>       | <code>\frontmatter</code>                              | DOC   | Opens the pre-textual part and typesets the approval sheet.                                                                                                                                                                              |
| <code>\mainmatter</code>        | <code>\mainmatter</code>                               | DOC   | Opens the textual part. This is what starts printing the folio.                                                                                                                                                                          |
| <code>\backmatter</code>        | <code>\backmatter</code>                               | DOC   | Opens the post-textual part.                                                                                                                                                                                                             |
| <code>\dedication</code>        | <code>\dedication{To those who waited.}</code>         | DOC   | Dedication, on a sheet of its own.                                                                                                                                                                                                       |
| <code>abstract</code>           | <code>\begin{abstract} ...</code>                      | DOC   | Abstract in the main language. One sheet, one paragraph, 250 words.                                                                                                                                                                      |
| <code>foreignabstract</code>    | <code>\begin{foreignabstract} ...</code>               | DOC   | Abstract in the foreign language.                                                                                                                                                                                                        |
| <code>brazilianabstract</code>  | <code>\begin{brazilianabstract} ...</code>             | DOC   | Third abstract, Portuguese, for a Spanish- or French-main work.                                                                                                                                                                          |
| <code>\configuraresumos</code>  | <code>\configuraresumos{true}{true}{true}{true}</code> | PRE   | Turns the four optional elements of every abstract sheet on or off, in every language: page identification, title and author, advising, reference. All on by default; only the abstract text itself cannot be turned off.                |
| <code>\appendix</code>          | <code>\appendix</code>                                 | DOC   | Opens the appendices: text <i>you</i> wrote.                                                                                                                                                                                             |
| <code>\annex</code>             | <code>\annex</code>                                    | DOC   | Opens the annexes: documents you did <i>not</i> write.                                                                                                                                                                                   |
| <code>\printbibliography</code> | <code>\printbibliography</code>                        | DOC   | The reference list, titled “Referências” (3.1.4.1).                                                                                                                                                                                      |
| <code>\coppetexfinalpage</code> | <code>\coppetexfinalpage</code>                        | DOC   | Optional colophon, every item measured at compile time: class version, engine and its version, TeX system, L <sup>A</sup> T <sub>E</sub> X format, font family and encoding, biblatex version, date, hour, operating system and machine. |

Only the table of contents is mandatory (3.1.2.1.6). Every list is optional (3.1.2.2); the COPPE norm (§10) adds the lists of *quadros*, programs and algorithms whenever the work has one of them.

#### PRE-TEXTUAL LISTS — lists first, table of contents last (3.1.2.1.6)

|                                    |                                    |     |                                                               |
|------------------------------------|------------------------------------|-----|---------------------------------------------------------------|
| <code>\listoffigures</code>        | <code>\listoffigures</code>        | DOC | List of figures.                                              |
| <code>\listoftables</code>         | <code>\listoftables</code>         | DOC | List of tables.                                               |
| <code>\listofquadros</code>        | <code>\listofquadros</code>        | DOC | List of <i>quadros</i> (framed on all sides, unlike a table). |
| <code>\listofprogramas</code>      | <code>\listofprogramas</code>      | DOC | List of code listings.                                        |
| <code>\listofalgorithms</code>     | <code>\listofalgorithms</code>     | DOC | List of algorithms.                                           |
| <code>\printloabbreviations</code> | <code>\printloabbreviations</code> | DOC | List of abbreviations and acronyms. No page numbers (4.1.1).  |
| <code>\printlosymbols</code>       | <code>\printlosymbols</code>       | DOC | List of symbols. Likewise.                                    |
| <code>\tableofcontents</code>      | <code>\tableofcontents</code>      | DOC | The table of contents. Comes last of all the lists.           |

#### ILLUSTRATIONS — caption on top, source below (2.10)

|                        |                                               |     |                                                           |
|------------------------|-----------------------------------------------|-----|-----------------------------------------------------------|
| <code>\source</code>   | <code>\source{Prepared by the author.}</code> | DOC | The source line, below the float. Mandatory by the norm.  |
| <code>\fonte</code>    | <code>\fonte{Elaboração própria.}</code>      | DOC | The same command, Portuguese name.                        |
| <code>\cpsource</code> | <code>\cpsource{...}</code>                   | DOC | Clash-safe alias, always available.                       |
| <code>\cpfonte</code>  | <code>\cpfonte{...}</code>                    | DOC | Likewise.                                                 |
| <code>quadro</code>    | <code>\begin{quadro}[ht] ...</code>           | DOC | A float framed on all sides. Behaves like <b>figure</b> . |

| Command                                               | Example                                               | Where | What it does                                                                                                                                                                                                                                         |
|-------------------------------------------------------|-------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>python</code>                                   | <code>\begin{python}[caption={...}]</code>            | DOC   | Python listing, syntax-highlighted, caption on top.                                                                                                                                                                                                  |
| <code>java, xml, html, prolog</code>                  | <code>\begin{java}[...] ...</code>                    | DOC   | The same, for those languages.                                                                                                                                                                                                                       |
| <code>\pythoninline</code>                            | <code>\pythoninline{f(x)}</code>                      | DOC   | Code inside a paragraph.                                                                                                                                                                                                                             |
| <code>\javainline</code>                              | <code>\javainline{new X()}</code>                     | DOC   | Likewise.                                                                                                                                                                                                                                            |
| <code>\pythonexternal</code>                          | <code>\pythonexternal{code.py}</code>                 | DOC   | Listing read from a file.                                                                                                                                                                                                                            |
| <code>\javaexternal</code>                            | <code>\javaexternal{Code.java}</code>                 | DOC   | Likewise.                                                                                                                                                                                                                                            |
| <code>\pythonstyle</code>                             | <code>\pythonstyle</code>                             | DOC   | Apply a language's style without opening the environment. Also <code>\javastyle</code> , <code>\xmlstyle</code> , <code>\htmlstyle</code> , <code>\prologstyle</code> .                                                                              |
| <b>QUOTATIONS, ACRONYMS, SYMBOLS</b>                  |                                                       |       |                                                                                                                                                                                                                                                      |
| <code>\citacao</code>                                 | <code>\citacao{up to three lines}</code>              | DOC   | Short quotation, inside the paragraph, in quotation marks.                                                                                                                                                                                           |
| <code>longquote</code>                                | <code>\begin{longquote} ...</code>                    | DOC   | Quotation over three lines: indented 4 cm past the margin, smaller, single spaced, no quotation marks (2.2b).                                                                                                                                        |
| <code>\abbrev</code>                                  | <code>\abbrev[IoT]{IoT}{Internet of Things}</code>    | DOC   | Register an acronym. Optional first argument is the sort key.                                                                                                                                                                                        |
| <code>\syml</code>                                    | <code>\syml{\${\alpha\$}}{Learning rate}</code>       | DOC   | Register a symbol. Same optional sort key.                                                                                                                                                                                                           |
| <code>\sigla</code>                                   | <code>\sigla{ufrj}</code>                             | DOC   | Full form on first use, acronym afterwards (2.8). <code>\sigla*</code> forces the full form.                                                                                                                                                         |
| <b>CITATIONS — biblatex, with the natbib layer on</b> |                                                       |       |                                                                                                                                                                                                                                                      |
| <code>\cite</code>                                    | <code>\cite{key}</code>                               | DOC   | (SURNAME, 2026), or [1] with numbers.                                                                                                                                                                                                                |
| <code>\citet</code>                                   | <code>\citet{key}</code>                              | DOC   | SURNAME (2026).                                                                                                                                                                                                                                      |
| <code>\citep</code>                                   | <code>\citep{key}</code>                              | DOC   | (SURNAME, 2026).                                                                                                                                                                                                                                     |
| <code>\textcite</code>                                | <code>\textcite{key}</code>                           | DOC   | Same as <code>\citet</code> , biblatex spelling.                                                                                                                                                                                                     |
| <code>\parencite</code>                               | <code>\parencite{key}</code>                          | DOC   | Same as <code>\citep</code> .                                                                                                                                                                                                                        |
| <code>\citeauthor</code>                              | <code>\citeauthor{key}</code>                         | DOC   | The author alone.                                                                                                                                                                                                                                    |
| <code>\citeyear</code>                                | <code>\citeyear{key}</code>                           | DOC   | The year alone.                                                                                                                                                                                                                                      |
| <code>\citetitle</code>                               | <code>\citetitle{key}</code>                          | DOC   | The title alone.                                                                                                                                                                                                                                     |
| <code>\cite[p. 45]</code>                             | <code>\cite[p. 45, our translation]{key}</code>       | DOC   | Page, and any note, in the optional argument.                                                                                                                                                                                                        |
| <code>\citepapud</code>                               | <code>\citepapud[45]{Original}{1953}{key-read}</code> | DOC   | Citation of a citation (4.1.2.2.1). <b>Three</b> mandatory arguments: the original author and year, which are printed as text, and the key of the work you actually read, which is the only one that enters the list. Also <code>\citetapud</code> . |
| <code>\nocite</code>                                  | <code>\nocite{key}</code>                             | DOC   | Into the list without citing in the text.                                                                                                                                                                                                            |

Full manual: `coppe.pdf`, in Portuguese. Complete worked example: `example.tex`. Problems: <https://github.com/COPPE-UFRJ/CoppeTeX/issues>